Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE-402103

Department of Electronics and Telecommunication Engineering

End Semester Examination

Subject: Digital Communication (EC 32) **Semester: VI**

Date: 17/05/2019 **Time: 3 hr.** **Marks: 60**

INSTUCTIONS:

1. Question 1 is compulsory and carries 10 marks whereas Question no 2 to Question no 7 carries 10 marks each.
2. Attempt any 5 Questions from Question 2 to Question 7
3. Illustrate the answers with neat sketches, diagram etc. wherever necessary.
4. Necessary data is given in the respective questions. If such data is not given it means its knowledge is a part of examination.
5. If some part or parameter is noticed to be missing, appropriate data may be assumed and should be mentioned clearly.

Q.1 Attempt any five following (Two marks each) **10 M**

A. Draw ASK waveform for digital data stream 10011101
B. Define Entropy.
C. A communication system consists of six messages with probabilities 1/8, 1/8, 1/8, 1/8, ¼, and ¼ respectively. Determine the entropy of the system.
D. A QAM modulator uses 4 different amplitudes and 16 different phase angles. How many numbers of bits it transmits for each symbol?
E. Find the relationship between a and b for the probability density function $f(x)=ae^{-b|x|}$, where a and b are any integers.
F. Draw ASK waveform for digital data stream 10011101

Q.2 A. Explain μ law companding and A law companding **5 M**
B. Explain block diagram of a single channel simplex PCM system **5 M**

Q.3 A. Draw the line encoding waveforms for the binary data 1 1 0 0 0 0 1 0 for the following types of line encoding formats: **5 M**
a. Bipolar RZ
b. Polar NRZ
c. Bipolar NRZ-AMI
B. Explain the fundamental of TDM system **5 M**

Q.4	A.	Explain the QPSK modulator with the help of neat block diagram	5 M
	B.	Compute and tabulate the results for bandwidth efficiency of M-ary PSK signals for M=2, 4, 8, 16, and 32	5 M
Q.5	A.	Prove that the mutual information of a channel is related to the joint entropy of the channel input and channel output by I{X;Y}=H(X)+H(Y)-H(X,Y)	5 M
	B.	Consider a binary symmetric channel which has probability of error, p=0.2. Assume that the bit rate is 1000 bits per second; find the rate of information transmission over the channel.	5 M
Q.6	A.	Explain in details block diagram of digital communication system.	5 M
	B.	Derive mathematical expression of QASK signal.	5 M
Q.7	A.	Consider five samples of a random variable, {X}={0.5, 0.8, 0.3, 0.01, 0.95}, having a Gaussian probability distribution with a mean of 0.5 and a variance of 1. a. What is the sample average or mean value of samples? b. Find the statistical average or the mean value. c. Find the variance	5 M
	B.	Explain Gaussian Probability Distribution and Uniform Probability Distribution	5 M

*******BEST OF LUCK*******